ELIJAH MATHEW

Aerospace Engineering Track

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SUMMARY

Aspiring aerospace engineer with hands-on exposure to satellite systems and defense technology through Johns Hopkins APL's Space Science Program. Strong academic record (3.7 GPA) paired with proven leadership through Civil Air Patrol and a track record of improving student outcomes as a STEM tutor. Self-starter with experience developing AI-driven investment tools and independent orbital mechanics research. Actively building technical foundations in CAD and Python.

EDUCATION

River Hill High School — High School Diploma, Clarksville, MD *Expected June 2029*

GPA: 3.7/4.0

Relevant Coursework: PLTW Introduction to Engineering Design (IED), Foundations of Technology, Spanish (College Credit)

PROJECTS

Orbital Mechanics Simulation — Independent Research*2026*

- Developed Python simulation using Poliastro library to analyze orbital period and velocity variations across five LEO altitudes (300–1100 km), demonstrating Kepler's Third Law and two-body orbital dynamics
- Implemented parametric study with numerical orbit propagation, computing period (90–106 min), velocity (7.1–7.7 km/s), and daily orbit count (13.6–15.9) for circular orbits
- Visualized results with Matplotlib using 2D trajectory plots and quantitative trend analysis; created portfolio-ready technical documentation in Jupyter Notebook format
- Analyzed trade-offs in satellite mission design relevant to Earth observation and communications satellites

"Slate" — AI Investing Assistant — Independent Development*2025–2026*

- Built algorithmic trading assistant analyzing billions of market data points to generate buy/sell signals for stocks
- Engineered automated trade execution workflow with push notifications via Telegram bot; only manual action was executing recommended trades
- Achieved portfolio growth from $10 to $1,000 over one year (10,000% return) through data-driven market predictions
- Demonstrated expertise in financial data analysis, algorithm development, and automated notification systems

Rocket Payload Design Challenge — PLTW IED Course*2024*

- Designed and modeled 3D-printable payload housing for small-scale scientific instruments using Autodesk Inventor
- Applied engineering design process including brainstorming, sketching, 3D modeling, and prototyping
- Conducted stress analysis simulations to optimize structural integrity while minimizing mass

Satellite Communication Simulation — APL Space Science Program*Summer 2024*

- Collaborated with team to simulate satellite-to-ground communication links in real-world aerospace research environment
- Worked directly with APL scientists and engineers on satellite systems, space exploration, and defense technologies
- Applied aerospace research methodology and engineering problem-solving processes through hands-on labs

EXPERIENCE

Academic Tutor — Best in Class Education Center, Clarksville, MD*Sept 2024 – Present*

- Deliver one-on-one tutoring in mathematics and core STEM subjects, strengthening comprehension and measurably improving test performance
- Design personalized lesson plans aligned to school curricula to target and close individual learning gaps
- Build students' confidence and independence in problem-solving by reinforcing foundational concepts and study strategies

Space Science Camp Participant — Johns Hopkins University Applied Physics Laboratory (APL), Laurel, MD*Summer 2024*

- Selected through a competitive application process for APL's 8th Grade Space Science Summer Program, covering real-world aerospace and physics applications
- Worked directly with APL scientists and engineers on satellite systems, space exploration, and defense technologies
- Applied aerospace research methodology and engineering problem-solving processes through hands-on labs and demonstrations

LEADERSHIP & ACTIVITIES

Cadet Member — Civil Air Patrol (CAP), Maryland Wing, Maryland*2023 – Present*

- Earned the Wright Brothers Award for completing CAP's foundational aerospace and leadership curriculum
- Train in aerospace education, emergency services, and leadership development within a structured, military-style program
- Develop discipline, teamwork, and mission-focused communication under a formal chain of command

Member — STEM Mentors, River Hill High School, Clarksville, MD*2023 – Present*

- Mentor peers and younger students to build interest and confidence in science and engineering
- Support outreach events connecting students with STEM career pathways

Varsity/JV Athlete — Track & Field, River Hill High School, Clarksville, MD*2024 – Present*

- Compete at the varsity/JV level while maintaining a 3.7 GPA, reflecting strong time management under a demanding dual commitment

AWARDS & HONORS

- Award of Excellence, Howard County Pledge Camp — recognized for leadership and character
- Wright Brothers Award, Civil Air Patrol — completed foundational aerospace and leadership curriculum
- Competitively Selected, Johns Hopkins APL Space Science Program (2024)

TECHNICAL SKILLS & INTERESTS

Engineering: PLTW IED (3D modeling, design process, prototyping), Foundations of Technology

Tools & Programming: Python (NumPy, Matplotlib, Poliastro, Astropy), CAD (Autodesk Inventor), Jupyter Notebook, Telegram API

Aerospace Interests: Space systems, satellite technology, orbital mechanics, defense aerospace, aeronautics

Languages: English (native), Spanish (intermediate – college credit)